

FleetSpark — User Manual

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Audience: Fleet managers, admins, drivers using FleetSpark

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Getting Started

Creating Your Account

1. Navigate to `https://fleetspark.syncsphere11c.com`
2. Click **"Sign Up"**
3. Enter your organization name, your name, email, and password
4. You'll be the **Admin** of your organization
5. You'll land on the Dashboard

Inviting Team Members

1. Go to **Settings** → **Users**
2. Click **"Invite User"**
3. Enter their email and select a role:
 - **Admin** — Full control, can manage users and billing
 - **Fleet Manager** — Can manage vehicles, drivers, and view all data
 - **Driver** — Can only see their assigned vehicle
 - **Viewer** — Read-only access to dashboard and reports
4. They'll receive an email invitation to set their password

First-Time Setup Checklist

- ☐ Create your account
 - ☐ Invite team members
 - ☐ Set up your organization profile (Settings → Organization)
 - ☐ Add vehicles
 - ☐ Add drivers
 - ☐ Assign drivers to vehicles
 - ☐ Set up GPS devices (or use demo mode)
 - ☐ Configure alert thresholds (Settings → Alerts)
 - ☐ Test the dashboard
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Dashboard

The dashboard is your home screen. It gives you a real-time overview of your entire fleet.

What You See

Summary Cards (top):

- **Total Vehicles** — All active vehicles in your fleet
- **Active Now** — Vehicles currently in transit (% of fleet)
- **Charging** — Vehicles currently plugged in
- **In Maintenance** — Vehicles scheduled for or undergoing maintenance
- **Alerts** — Unacknowledged alerts requiring attention

Fleet Map:

- Each vehicle appears as a colored dot
- **Green** = In Transit | **Blue** = Charging | **Gray** = Idle
- **Red** = Has an alert | **Purple** = Maintenance | **Dark** = Offline
- Click any dot to see vehicle details
- Use the legend to understand colors

Alert Panel:

- Shows the latest unacknowledged alerts
- Click "**View All**" to see the full alerts page
- Red = Critical, Orange = Yellow = Warning, Blue = Info

Vehicle List:

- Table of all vehicles with key info
- Search by name or plate number
- Filter by status (transit, charging, idle, etc.)
- Click any row to open the Vehicle Detail page

Understanding Vehicle Status

Status	Icon	What It Means
In Transit	□	Vehicle is moving
Charging	□	Vehicle is plugged in and charging
Idle	○	Vehicle is parked, not charging
Maintenance	□	Vehicle is in the workshop
Alert	□	Vehicle has an active alert
Offline	●	No GPS signal received in 30+ minutes

Managing Vehicles

Adding a Vehicle

1. Click "**Vehicles**" in the sidebar
2. Click "**Add Vehicle**" button
3. Fill in the required fields:
 - **Name** — e.g., "Kayoola-001" (your internal name)
 - **Plate Number** — e.g., "UAX 123A"
 - **VIN** — Vehicle Identification Number (optional but recommended)
 - **Model** — e.g., "Kayoola EVS"
 - **Year** — e.g., 2025
 - **Type** — Select from: Bus, Coach, Car, Van, Truck
 - **Battery Capacity (kWh)** — e.g., 250
 - **Max Range (km)** — e.g., 300
 - **Depot** — Where the vehicle is based (optional)
4. Click "**Save**"

Editing a Vehicle

1. Click on the vehicle row in the list
2. Go to the "**Settings**" tab
3. Click "**Edit**"
4. Update any field
5. Click "**Save**"

Assigning a GPS Device

1. In Vehicle Settings, find the **Device ID** field
2. Enter the ID of your GPS device (e.g., "teltonika-abc123")
3. Make sure the device is already sending telemetry (see [GPS Device Setup](#))
4. Once telemetry is received, the vehicle will appear on the map

Deactivating a Vehicle

1. In Vehicle Settings, toggle **"Active"** to off
2. The vehicle will be hidden from the main list but data is preserved
3. To reactivate, toggle back on

Vehicle Detail Page

Click any vehicle to see its detail page with tabs:

Overview Tab:

- Current location on mini-map
- Current battery level, SoH, estimated range
- Today's trips summary
- Assigned driver info
- Vehicle info (odometer, battery capacity, etc.)

Location Tab:

- Full-size map with the vehicle's current position
- Route trail showing where it's been (last 1 hour by default)
- Change time range: Last 1h, 6h, 24h, 7 days
- Geofence status (if configured)

Battery Tab:

- Large battery gauge showing current level
- State of Health (SoH) percentage
- Estimated remaining range
- Battery history chart (last 24h, 7 days, 30 days, 90 days)
- Charging sessions table with dates, levels, and energy consumed

Trips Tab:

- List of all trips with date, distance, duration, avg speed
- Click a trip to see the route on a map
- Filter by date range

Maintenance Tab:

- Upcoming maintenance (next 30 days)
- Maintenance history
- **"Schedule Maintenance"** button to add new

Settings Tab:

- Edit vehicle info
- Configure max range (for accurate range estimates)
- Configure alert thresholds for this specific vehicle

Managing Bikes

Bikes (e-bodas, e-motorcycles) are managed similarly to vehicles but with bike-specific fields.

Adding a Bike

1. Click **"Vehicles"** → **"Add Vehicle"**
2. Select **Type: Bike** (or "Motorcycle")
3. Fill in:
 - **Name** — e.g., "Spiro-001"

- **Plate Number** — e.g., "UAX 500M"
- **Model** — e.g., "Spiro Max"
- **Battery Capacity (kWh)** — e.g., 2.5 (much smaller than buses)
- **Max Range (km)** — e.g., 80

4. Click "**Save**"

Bike-Specific Features

- **Battery swap tracking:** FleetSpark detects when a battery swap occurs (sudden level jump from low to high)
- **Swappable battery ID:** Optionally assign a battery ID to track individual batteries
- **Smaller battery gauge:** UI adapts to show appropriate range for bikes
- **Bike on map:** Shown with a motorcycle icon (🏍️) vs bus icon (🚌)

Managing a Bike Fleet

Bikes appear alongside vehicles in the dashboard. Use the **Type filter** to show only bikes.

For large bike fleets (100+):

- Use the **search** function to find specific bikes
- **Bulk import** via CSV (admin feature in v2)
- Assign **battery swap stations** as sub-locations

Battery Swap Detection

FleetSpark automatically detects battery swaps:

1. Battery level drops to <10%
2. Suddenly jumps to >90% within 5 minutes (without charging status)
3. System logs a "Battery Swap" event
4. Visible in the Battery tab as a swap marker

Managing Drivers

Adding a Driver

1. Click "**Drivers**" in the sidebar
2. Click "**Add Driver**"
3. Fill in:
 - **Name** — Full name
 - **Phone** — With country code (e.g., +256...)
 - **Email** — Optional
 - **License Number** — Optional
 - **License Expiry Date** — Optional
4. Click "**Save**"

Assigning a Driver to a Vehicle

1. Open the **Driver Detail** page
2. Click "**Assign to Vehicle**"
3. Select the vehicle from the dropdown
4. Click "**Confirm**"

5. The driver will now appear on the vehicle's detail page

Reassigning a Driver

1. Open the **Vehicle Detail** page
2. Click **"Change Driver"** in the driver section
3. Select a new driver or choose **"Unassign"**
4. Click **"Confirm"**

Driver View

When a driver logs in:

- They only see their assigned vehicle(s)
 - They can see current battery level and estimated range
 - They can update their location manually (if no GPS device)
 - They cannot see other vehicles or fleet analytics
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Battery Monitoring

Understanding Battery Level

- **Gauge:** Visual indicator from 0% to 100%
- **Color coded:** Green (>50%), Yellow (20-50%), Red (<20%)
- **Estimated Range:** Calculated from battery level × max range

Understanding State of Health (SoH)

- **SoH** measures battery degradation over time
- **100%** = Brand new battery
- **80%** = Battery has degraded 20% (needs monitoring)
- **<70%** = Battery replacement recommended
- SoH naturally decreases with charge cycles

Charging Sessions

A charging session is recorded when:

1. Vehicle reports `charging: true`
2. Vehicle reports `charging: false`
3. System calculates: duration, start/end level, estimated energy consumed

Low Battery Alerts

- Default threshold: **20%**
- Configurable in **Settings → Alerts**
- Notifications: In-app, email, or SMS
- Acknowledge the alert to clear it from the active list

Battery History Chart

- View battery level over time
 - Toggle: 24 hours, 7 days, 30 days, 90 days
 - Charging periods highlighted in blue
 - Useful for identifying battery degradation patterns
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Trip History

What is a Trip?

A trip is automatically recorded when:

1. The vehicle starts moving (ignition on / speed > 0)
2. The vehicle stops for more than 5 minutes (ignition off)

Trip Data

- **Start/End time** — When the trip began and ended
- **Start/End location** — GPS coordinates and reverse-geocoded address
- **Distance** — Kilometers covered (from GPS)
- **Duration** — Total time in minutes
- **Average speed** — Mean speed during the trip
- **Maximum speed** — Highest speed recorded

Viewing Trip Route

1. Open a trip from the list
2. Click "**Show Route**"
3. The map displays the exact path taken
4. Colored by speed (green = fast, red = slow)

Export Trips

1. Go to **Reports**
 2. Select **Trip Export**
 3. Choose date range
 4. Click "**Export CSV**"
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Maintenance Scheduling

Creating a Maintenance Record

1. Open the **Vehicle Detail** → **Maintenance** tab
2. Click "**Schedule Maintenance**"
3. Fill in:
 - **Type:** Routine, Repair, Inspection, Battery Service, Tire, Brake, Other

- **Description:** What needs to be done
 - **Scheduled Date:** When it should happen
 - **Odometer:** Current reading (optional)
4. Click "**Save**"

Completing Maintenance

1. Open the maintenance record that's due
2. Click "**Mark Complete**"
3. Enter:
 - **Completed Date:** When it was done
 - **Cost (UGX):** How much it cost
 - **Notes:** What was done, parts replaced, etc.
4. Click "**Save**"

Maintenance Alerts

- System generates an alert when maintenance is due
- Default: 7 days before scheduled date
- Configurable in [Settings](#) → [Alerts](#)

Alerts & Notifications

Alert Types

Alert	Description	Default Trigger
Low Battery	Battery below threshold	< 20%
Offline	No GPS signal received	> 30 minutes
Maintenance Due	Scheduled maintenance approaching	7 days before
Geofence Exit	Vehicle left designated area	On exit
High Temperature	Battery overheating	> 45°C
Rapid Battery Drop	Unexpected battery drain	> 10% in 15 min

Acknowledging Alerts

1. Click the **alert** in the panel or alerts page
2. Click "**Acknowledge**"
3. The alert moves to "Acknowledged" status
4. You can add a note explaining action taken

Configuring Alert Thresholds

1. Go to [Settings](#) → [Alerts](#)
2. For each alert type, adjust:
 - **Threshold** (e.g., change low battery from 20% to 25%)

- **Email notifications** (on/off)
- **SMS notifications** (on/off and enter phone number)
- **In-app notifications** (on/off)

3. Click "**Save**"

SMS Notifications

- SMS uses Twilio (requires balance)
 - Standard SMS rates apply (~UGX 150/SMS)
 - Configure your phone number in **Settings** → **Organization**
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Reports

Fleet Summary Report

Shows fleet-wide metrics for a date range:

- Total distance covered (km)
- Total trips
- Average fleet utilization (%)
- Total energy consumed (kwh)
- Total charging sessions
- Cost savings vs diesel

Cost Savings Report

Calculates how much money you've saved by using EVs:

- **Fuel saved** = (Distance covered ÷ Average diesel km/liter)
- **Cost saved** = Fuel saved × Diesel price per liter (default: UGX 5,500)
- **CO₂ saved** = Fuel saved × Emission factor (default: 2.68 kg/liter)

Vehicle Utilization Report

Per vehicle:

- Total distance (km)
- Number of trips
- Utilization % (time in transit vs idle)
- Energy consumed (kWh)
- Cost savings

Exporting Reports

1. Select the report type
 2. Set date range
 3. Click "**Export PDF**" (formatted document) or "**Export CSV**" (raw data)
 4. The file downloads to your device
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Settings

Organization Settings

- **Name:** Your company/organization name
- **Address:** Physical address
- **Phone:** Contact number
- **Logo:** Upload your company logo
- **Default Diesel Price:** Used for cost savings calculations
- **Default Emission Factor:** Used for CO₂ calculations
- **Timezone:** Set to "Africa/Kampala" for Uganda

User Management

- See all users in your organization
- Invite new users
- Change user roles
- Deactivate users

Alert Configuration

- Set thresholds for each alert type
- Configure notification channels (email/in-app/SMS)

Telemetry API Keys

- Generate API keys for GPS devices
 - Each key is linked to your organization
 - Devices use this key to send telemetry data
 - If compromised, revoke and generate a new key
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Roles & Permissions

Action	Admin	Fleet Manager	Driver	Viewer
View dashboard	☐	☐	Own vehicle only	☐
Add/edit vehicles	☐	☐	☐	☐
Delete vehicles	☐	☐	☐	☐
Add/edit drivers	☐	☐	☐	☐
Assign drivers	☐	☐	☐	☐
View battery data	☐	☐	Own vehicle only	☐
Acknowledge alerts	☐	☐	☐	☐
Configure alerts	☐	☐	☐	☐
View reports	☐	☐	☐	☐
Export data	☐	☐	☐	☐
Manage users	☐	☐	☐	☐
Manage billing	☐	☐	☐	☐

GPS Device Setup

Supported Devices

- **Teltonika FMB920** (recommended for vehicles)
- **Teltonika FMB120** (with CAN bus support)
- **Generic OBD-II** telematics devices that support HTTP POST
- **Smartphone app** (for manual location updates)

Configuration Steps

1. Generate a **Telemetry API Key** in FleetSpark (Settings → Telemetry Keys)
2. Assign the device to a vehicle (Vehicle Settings → Device ID)
3. Configure the GPS device to send HTTP POST to:
POST `https://fleetspark.syncspherellc.com/api/v1/telemetry` Headers: X-API-Key: <your_api_key>
Body: { "deviceId": "...", "timestamp": "...", "gps": {...}, "battery": {...} }
4. Set telemetry interval to **30 seconds**
5. Verify: The vehicle should appear on the map within 1-2 minutes

Demo / Testing Mode

If you don't have GPS hardware yet:

1. Go to **Settings → Demo Mode**
2. Click **"Generate Demo Data"**

3. System creates 5 simulated vehicles with fake GPS movement
4. Useful for testing the dashboard before hardware arrives

FAQ

Q: How often does the location update?

A: Every 30 seconds (configurable on the GPS device).

Q: What happens if internet is down?

A: The GPS device buffers data and sends reconnection. FleetSpark stores last known location and shows "Offline" status.

Q: Can I track both EVs and diesel vehicles?

A: Yes! FleetSpark works with any vehicle type. For non-EVs, battery features are simply not shown. For EVs, you also get charging and cost savings.

Q: Can my drivers use their phones instead of GPS hardware?

A: Yes. The mobile web app allows drivers to manually update location. Accuracy depends on phone GPS.

Q: How is cost savings calculated?

A: $(\text{Distance covered in km}) \div (\text{your diesel vehicle's km/liter}) \times (\text{diesel price per liter})$. Set your diesel price in Settings → Organization.

Q: Can I see historical data from before I started using FleetSpark?

A: No. FleetSpark only stores data from the moment you start sending telemetry. For historical data, export from your old system and import via CSV (v2 feature).

Q: How many vehicles can I track?

A: Up to 200 per organization on the Starter plan. Enterprise plans support unlimited vehicles.

Q: Is my data secure?

A: Yes. All data is encrypted in transit (HTTPS) and at rest. Each organization's data is isolated. We never share your data.

Q: Can I white-label FleetSpark with my own branding?

A: Yes, on the Enterprise plan. Contact us for pricing.

Support

- **Email:** support@fleetspark.syncsphereinc.com
- **Phone:** +1 (432) 692-0996
- **Hours:** Monday – Friday, 9AM – 6PM EAT
- **Emergency:** For critical issues (system down), call the emergency line

FleetSpark v1.0 — Built by SyncSphere LLC